Department of Biotechnology, School of Energy Technology

B.Tech Biotechnology

Course Matrix for Semester – IV (Batch 2024-2028)

Semester	Category Code	Course Code	Course Name	Theory	Tutorial	Practical	Hrs	Credits
Semester 4	ESC	24BT206T	Industry 4.0 (Department Specific)	2	0	0	2	2
	ESC	24BT206P	Industry 4.0 - Lab (Department Specific)	0	0	2	2	1
	PC	24BT207T	Molecular Biology and Genetics	3	0	0	3	3
	PC	24BT207P	Molecular Biology & Genetics Laboratory	0	0	2	2	1
	PC	24BT208T	Bioprocess Engineering	3	0	0	3	3
	PC	24BT208P	Bioprocess Engineering Lab	0	0	2	2	1
	PC	24BT209T	Human Anatomy and Physiology - II	3	0	0	3	3
	PC	24BT209P	Human Anatomy and physiology Lab -II	0	0	2	2	1
	PC	24BT210T	Analytical Bioinformatics	3	0	0	3	3
	PC	24BT210P	Analytical Bioinformatics Lab	0	0	2	2	1
	OE	24BT221T	Biomedical Informatics	3	0	0	3	3
	Pro	24INT251	Industrial Orientation	0	0	0	0	0

24BT207T					Molecular Biology and Genetics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	50	50	100

COURSE OBJECTIVES

1. The course aims to give an understanding on the fundamentals of molecular biology
2. To understand the fundamentals of DNA repair mechanisms
3. Explore the principles of genetics and inheritance patterns
4. Understand the molecular basis of genetic modifications

UNIT I: INTRODUCTION 12 Hrs.

History of molecular biology; Central dogma of life; Structure of DNA and RNA; Gene structure and function; DNA replication; transcription; Genetic code and translation; Structure of prokaryotic and eukaryotic nuclear and organelle genomes; Gene regulation in prokaryotes: Lac operon concept, trp concept.

UNIT II: MICROBIAL PATHOGENICITY 12 Hrs.

Post translational modification, DNA repair and Recombination, DNA repair mechanisms and their types, Holliday Junction, Process of recombination. Programmed cell death, cell junctions and adhesion.

UNIT III: GENETICS 8 Hrs.

Mendelian genetics and heredity, Mendel's experiments, principles of segregation, Principle of independent assortment, polygenic inheritance, Multiple alleles, Linkage and Crossing Over, Eugenics

UNIT IV: MUTATION AND MICROBIAL GENETICS 10 Hrs.

Molecular basis of mutation and their different types, Microbial genetics: conjugation, transformation, transduction, Plasmids and transposable elements, Chromosomal variation; Genetic disorders; Population genetics; Epigenetics; Selection and inheritance; Adaptive and neutral evolution; Genetic drift; Species and speciation

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Explain the foundations of Mendelian genetics and chromosomal theory and apply these, with appropriate terminology, to contemporary concepts in genetics.
CO2 : Emphasize the molecular mechanism of DNA replication and repair in various organisms
CO3 : Explain the properties of genetic materials and storage and processing of genetic information.
CO4 : Analyse the processes of transcription and translation in both prokaryotes and eukaryotes at molecular level.
CO5 : Understand the redundant and universal qualities of the genetic code and how it is used to determine the amino acid sequence of a polypeptide.
CO6 : Compare the mechanisms of gene regulation in prokaryotes and eukaryotes.

TEXT / REFERENCE BOOKS

1. Jocelyn Krebs, Stephen Kilpatrick and Elliott Goldstein, "Lewin B., Genes XI", International Edition, Jones & Bartlett Learning, 2017, ISBN 978-1-4496-5985-1
2. Bruce Alberts, Alexander Johnson, Julian Lewis, David Morgan, Martin Raff, Keith Roberts "Molecular Biology of the Cell", W.W. Norton & Company, ISBN: 0815344643
3. Harvey Lodish, Arnold Berk, Chris Kaiser, Monty Krieger, Matthew Scott, Anthony Bretscher, Hidde Ploegh, Paul Matsudaira, "Molecular Cell Biology" by W.H. Freeman publisher, ISBN: 142920314
4. Pierce, B.A. (2012) "Genetics - A Conceptual Approach", (6h ed.), W.H. Freeman & Co. (New York), ISBN: 13:978-1-4292-7606-1 / ISBN: 10:1-4292-7606-1.
5. D. Peter Snustad, Michael J. Simmons, "Principles of Genetics", 7th Edition, Wiley 2015.

24BT207P					Molecular Biology & Genetics Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

1. To provide students the knowledge about gene organization, genetic materials, molecular heritance, gene transfer, and their regulations.
2. To provide practical skills in molecular biology techniques
3. To develop critical thinking and analytical skill to interpret molecular biology data
4. To emphasize the importance of experimental design and data interpretation

LIST OF EXPERIMENTS

1 Making competent *E. coli* cells using chemical methods.
2 Transformation of *E. coli* and calculation of competence.
3 Extraction of plasmid DNA and calculation of concentration and purity.
4 Extraction of genomic DNA from animal and plant tissues and calculation of concentration and purity.
5 Estimation of size in bp of DNA using agarose gel electrophoresis.
6 Restriction mapping of pUC19 vector.
7 Extraction of proteins from plant or animal tissue and confirmation with qualitative tests
8 Separation and identification of proteins by SDS-PAGE using Coomassie Brilliant Blue stain
9 Extraction of RNA from animal and plant tissues
10 Preparation of Drosophila polytene chromosome squashes

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Gain fundamental knowledge on electrophoresis techniques.
CO2 : Get acquainted with calculation of concentration and purity.
CO3 : Get acquainted with extraction protocol
CO4 : Analyze experimental data obtained from molecular biology experiments
CO5 : Evaluate, troubleshoot and validate the molecular biology experiments
CO6 : Design experiments incorporating fundamentals of molecular biology.

TEXT/REFERENCE BOOKS:

1. Jocelyn Krebs, Stephen Kilpatrick and Elliott Goldstein, “Lewin B., Genes XI”, International Edition, Jones & Bartlett Learning, 2017, ISBN 978-1-4496-5985-1
2. D. N. Bharadwaj “Textbook on Molecular Genetics” (2009); Kalyani Publisher
3. Ausubel, F.M. et al. (2012), “Current protocols in molecular biology”, New York: John Wiley & Sons.
4. Green, M. R., & Sambrook, J. (2012). Molecular cloning: A laboratory manual (4th ed., Vol. 1-3), Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press.

24BT208T					Bioprocess Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	50	50	100

COURSE OBJECTIVES

1. Learn the necessary basic protocols of cultivation, growth kinetics of microorganism, sterilization.
2. Understand the industrial applications of bioprocess engineering.
3. Gain the fundamental on the media formulation and scale up in bioprocess.
4. Create interest in learning bioreactor design and process control in bioprocess engineering system.

UNIT I: INTRODUCTION 10 Hrs.

Outline of an integrated bioprocess: Upstream and downstream, Unit operations involved in bioprocesses, Process flow sheets of metabolite production, Microbial kinetics of growth, Modes of bioreactor operation: Batch; Fed batch and Continuous operation, Production kinetics, Kinetics of substrate uptake and Yields in cell culture.

UNIT II: MEDIA DESIGN AND STERILIZATION KINETICS 10 Hrs.

Type of Sterilizations, Design of batch heat sterilization process, Design of continuous heat sterilization process, Filter sterilization of fermentation media, Medium formulation: Carbon sources, Nitrogen sources, Minerals, Chelators, Growth factors, Antifoams, Addition of precursors and Metabolic regulators to media, Introduction to media optimization. .

UNIT III: DESIGN OF BIOREACTORS 10 Hrs.

Design and components of various bioreactors: Stirred tank (CSTR); Packed bed; Bubble-column; Fluidized bed; Trickle bed; Airlift loop; Photo bioreactors. Bioreactor control mechanism, Physical, Chemical and Biological environment of bioreactor, Role of physical, chemical & biological sensors, Online and offline analytical methods.

UNIT IV: APPLICATIONS OF BIOPROCESS ENGINEERING 10 Hrs.

Food Industry (Baker Yeast Production), Environmental Industry (Biological Waste Water Treatment), Medical Applications of Bioprocess Engineering (Tissue Engineering, Gene Therapy).

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Classify the functions of a bioreactor and various types of bioreactors.
CO2 : Gain the fundamental knowledge on sterilization kinetics.
CO3 : Understand about the industrial fermentation process and process flow sheet.
CO4 : List media requirements and examine the formulation and optimization for fermentation.
CO5 : Focus professionally to select the right ancillaries or equipment's for the industrial units
CO6 : Develop solution scale up and scale down of bioprocess system.

TEXT/REFERENCE BOOKS

1. Stanbury P., Whitaker A and Hall S, "Principles of Fermentation Technology", 2016, 3rd Edition, Science & Technology Books.
2. Doran P.M. "Bioprocess Engineering Principles", 2012, 2nd Edition, Academic Press.
3. Shuler M.L and Kargi F. "Bioprocess Engineering: Basic Concepts", 2012, 2nd Edition, Prentice Hall Inc.
4. Bailey J.E and Ollis D.F. "Biochemical Engineering Fundamentals", 2007, 2nd Edition, McGraw Hill.

24BT208P					Bioprocess Engineering Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. Experimentation with different ideal reactors for a homogeneous liquid phase reactions to determine kinetics
2. To find the kinetic parameter using Arrhenius theory by experimentation with different ideal reactors
3. Find the application of combination of ideal reactors and the change in performance with the ideal reactors
4. To find the non-catalytic homogeneous liquid phase reaction

LIST OF EXPERIMENTS

1 Determine the growth patterns of Escherichia coli
2 Determination of specific thermal death rate constant (kd) for Escherichia coli
3 To analyse the effect of nutritional inhibitors on bacterial growth
4 Upstream of bioprocess for the production of Citric acid by Aspergillus niger
5 Bioprocess for the production of enzymes by Aspergillus oryzae
6 Preparation of immobilized enzymes & cells and evaluation of kinetic parameters
7 Ethanol fermentation using Saccharomyces cerevisiae
8 To study a non-catalytic homogeneous liquid phase reaction in an ambient CSTR
9 To study a non-catalytic homogeneous liquid phase reaction in an ambient straight tube PFR
10 To study a non-catalytic homogeneous liquid phase reaction in an ambient semi Batch reactor
11 To study a non-catalytic homogeneous liquid phase reaction in an ambient three Cascaded CSTR
12 Evaluating kinetic parameters (k0 and E) for a saponification reaction in an Batch Reactor
13 Evaluating kinetic parameters (k0 and E) for a saponification reaction in an Coiled PFR
14 Evaluating kinetic parameters (k0 and E) for a saponification reaction in an CSTR
15 To study a non-catalytic homogeneous liquid phase reaction in an ambient combined flow reactor (CSTR+PFR)

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Relate to the basics of kinetics and application of the same in practical
CO2 : Interpret and evaluate the rate data and get the kinetics parameters for different reactors
CO3 : Apply proper rate equations and compare the performance of different ideal reactors
CO4 : Examine the combinations of ideal reactors and analyze the theoretical findings
CO5 : Decide the reactor sequencing so as to optimize the volume or conversion for given combinations of reactors
CO6 : Develop an ability to work in group, learn report making, and improve soft skills in representing the findings

TEXT/REFERENCE BOOKS

1. Stanbury P., Whitaker A and Hall S, “Principles of Fermentation Technology”, 2016, 3rd Edition, Science & Technology Books.
2. Doran P.M. “Bioprocess Engineering Principles”, 2012, 2nd Edition, Academic Press.
3. Shuler M.L and Kargi F. “Bioprocess Engineering: Basic Concepts”, 2012, 2nd Edition, Prentice Hall Inc.
4. Bailey J.E and Ollis D.F. “Biochemical Engineering Fundamentals”, 2007, 2nd Edition, McGraw Hill.

24BT209T					Human Anatomy and physiology -II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	50	50	100

COURSE OBJECTIVES

1. To gain the fundamental knowledge on Structure and functions of the various organs and systems of the human body.
2. To learn the skeletal architecture and significance of joint articulation.
3. To understand the vital functions of Digestive, Urinary, Endocrine and reproductive system.
4. To evaluate the physiological aspect of Endocrine hormones and significance of reproduction system.

UNIT I: DIGESTIVE AND URINARY SYSTEM 10 Hrs.

Anatomy and Physiology of the GIT with special reference to associated organs and accessory glands. Physiology of digestion and absorption of Nutrients. disorders of GIT. Anatomy and physiology of urinary system. Physiology of urine formation. Mechanism of micturition. Regulation of Renin - angiotensin system.

UNIT II: SKELETAL SYSTEM AND JOINTS 12 Hrs.

Divisions of skeletal system, types of bone, salient features and functions of bones of axial and appendicular skeletal system, Organization of skeletal muscle, physiology of muscle contraction, neuromuscular junction. Structural and functional classification, types of joints movements and its articulation

UNIT III: ENDOCRINE SYSTEM 10 Hrs.

Classification of hormones, structure and functions of pituitary gland, parathyroid gland, thyroid gland, adrenal gland, pancreas, thymus and their disorders

UNIT IV: REPRODUCTIVE SYSTEM 10 Hrs.

Anatomy of male and female reproductive system, Functions of male and female reproductive system, sex hormones, physiology of menstruation, fertilization.

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Understand the anatomy of Skeletal system, Joints, Digestive, Excretory, Endocrine and reproductive system

CO2 : Demonstrate the involvement of vital organs, bones, hormones and their control mechanism.

CO3 : Classify and compare various bones, hormones, digestive mechanism, reproductive pathways, organs and their applications.

CO4 : Get acquainted with functions of Joints, GIT, Excretory Organs and hormones of human body.

CO5 : Focus on consequence of various disease conditions pertaining to Joints, Digestive, Excretory, Endocrine and reproductive system.

CO6 : Design and develop solutions to various clinical conditions.

TEXT/REFERENCE BOOKS

1. Ross and Wilson," Anatomy and Physiology in Health and illness", Elsevier.
2. K. Sambulingam and P Sambulingam," Fundamentals of Medical Physiology", Jaypee Brothers Medical Publishers.
3. S. Chaudhary and A. Chaudhary," Human Anatomy and Physiology", S Vikas and Company.
4. C. C. Chatterjee," Human Physiology", CBS Publisher.
5. Goyal R.K., Natvar M.P. and Shah S.A," Practical Anatomy, Physiology and Biochemistry, Experimental Physiology",B.S.Books.
6. Arthur C, Guyton and John.E," Text book of Medical Physiology", Elsevier.

24BT209P					Human Anatomy and physiology Lab -II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To understand and relate the concepts learned in Human anatomy and physiology to the real-world situations.
2. To acquire skills to perform laboratory experiments.
3. To analyses and compare different hematology parameters.
4. To analyses and compare different Serology parameters.

LIST OF EXPERIMENTS

1 Microscopic study of organs with slides
2 Identification of system with Specimens, charts & models
3 Estimation of hemoglobin content
4 Demonstration of serology parameters by auto analyzer
5 Determination of Blood Glucose level
6 Determination of Serum Cholesterol level
7 Determination of SGOT level
8 Determination of SGPT level
9 Determination of Blood urea level
10 Determination of Creatinine level
11 Demonstration of total blood count by hematology analyzer
12 Enumeration of white blood cell (WBC) count
13 Enumeration of total red blood corpuscles (RBC) count.
14 Demonstration on Mechanical ventilator

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Study the fundamentals concepts involved in performing laboratory experiments in physiology
CO2 : Acquire skills to perform laboratory experiments in physiology.
CO3 : Apply the standard laboratory protocols in clinical assessment
CO4 : Analyse the applications of haematology and serology profiling using standard operating procedures
CO5 : Evaluate the clinical standards of different physiological parameters during pathological condition
CO6 : Design and develop standard laboratory protocols for assessment of clinical conditions

TEXT/REFERENCE BOOKS

1. Ross and Wilson," Anatomy and Physiology in Health and illness", Elsevier.
2. Inderbir Singh," Textbook of Human Histology", Jaypee Brothers Medical Publishers.
3. C.L. Ghai," Human Anatomy and Physiology", Jaypee Brothers Medical Publishers.
4. K. Srinageswari and Rajeev Sharma," Practical workbook of Human Physiology", Jaypee Brothers Medical Publishers.
5. Goyal R.K., Natvar M.P. and Shah S.A," Practical Anatomy, Physiology and Biochemistry, Experimental Physiology",B.S.Books.

24BT210T					Analytical Bioinformatics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	50	50	100

COURSE OBJECTIVES

1. Learn the necessary basic protocols in computers and various biological databases.
2. Understand the basic essential tools in bioinformatics and implementation.
3. Gain the fundamental on the tools used for biological sequential data analysis.
4. Create interest in Phylogenetic analysis and DNA mapping and sequencing.
5. Learn the concepts of analysing genetic and protein information.

UNIT I: INTRODUCTION 10 Hrs.

Scope of Bioinformatics – Elementary commands and Protocols, ftp, telnet, http. Databanks – nucleotide databanks – Genbank, NCBI, EMBL, DDBJ – protein databanks – sequence databanks – PIR, SWISSPROT, TrEMBL - structural databases – PDB, SCOP, CATH.

UNIT II: SEQUENCE ALIGNMENT AND DYNAMIC PROGRAMMING 10 Hrs.

Introduction – Strings – Edit distance two strings – string similarity - local alignment - gaps – Parametric sequence alignments – suboptimal alignments – multiple alignment – common multiple alignment methods.

UNIT III: SEQUENCE DATABASES, MAPPING AND ALIGNMENT 10 Hrs.

Database search – Algorithms issues in database search – sequence database search – FASTA – BLAST – Amino acid substitution matrices PAM250 and BLOSUM62. Mapping and genome project – Physical and genetic maps - Map alignment – Sequence assembly.

UNIT IV: MOLECULAR PREDICTION 10 Hrs.

Gene prediction methods and evaluation – Gene prediction in microbial genome and eukaryotes - Molecular predictions with DNA strings – Protein secondary structure prediction methods, Molecular Docking methods.

40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Classify and Compare the different biological databases
CO2 : Gain the fundamental knowledge on sequence alignment and mapping
CO3 : Understand the principles of BLAST and FASTA
CO4 : Get acquainted with the algorithms used in sequence and structure databases
CO5 : Focus professionally on software's and tools used for nucleotide and protein understanding
CO6 : Develop solution for different prediction of protein structure and molecular docking methods.

TEXT/REFERENCE BOOKS

1. Dan Gusfield, "Algorithms on Strings Trees and Sequences", Cambridge University Press, Cambridge, 2022.
2. David Mount W., "Bioinformatics sequence and genome analysis", CBS Publishers, New Delhi, 2nd Edition, 2019.
3. Attwood T. K. and Parry-Smith D., "Introduction to Bioinformatics", Pearson Education, Delhi, 2021.
4. Arthur Lesk M, "Introduction to Bioinformatics", Oxford University Press, New Delhi, 2018.

24BT210P					Analytical Bioinformatics Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To view and use the various biological databases available on the World Wide Web.
2. To retrieve the gene sequence in FASTA format
3. To identify the Genes present if any in the given genomic sequence
4. To predict primary, secondary and tertiary structure of the give protein sequences

LIST OF EXPERIMENTS

1 Exploration of the resources available in NCBI and PUBMED
2 Retrieval of a Genbank Entry using an accession number
3 Retrieval and analysis of a gene sequence in FASTA format
4 Finding the official Symbol, alias name, chromosome number and ID for gene using NCBI
5 Retrieval and analysis of a protein sequence from protein database
6 Primary structure analysis of a protein
7 Secondary structure analysis of a protein
8 Tertiary protein structure analysis using RASMOL
9 Pair-wise and multiple sequence alignment using ClustalW
10 Pair-wise and multiple sequence alignment using BLAST

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Retrieve a nucleotide sequence of interest from Genbank and NCBI database
CO2 : Analyse the NCBI web site and find the official gene symbol, its alias name, chromosome number and its ID
CO3 : Analyse and retrieve the protein sequence of a protein from the Protein Data Bank (PDB) database
CO4 : Explore the site map of NCBI and PUBMED and to study the resources available on NCBI and PUBMED
CO5 : Perform pair wise and multiple sequence alignment using clustalw for given sequences
CO6 : Develop solution for different prediction of protein structure and molecular docking methods.

TEXT/REFERENCE BOOKS

1. Orpita Bosu, “Bioinformatics: Experiments, Tools, Databases, and Algorithms (Oxford Higher Education)”, 2017.
2. Dev Bukhsh Singh (Editor), Rajesh Kumar Pathak, “Bioinformatics: Methods and Applications”, 1st Edition. 2019
3. Jean-Louis Lassez, “Introduction to Bioinformatics Using Action Labs”, 2020

24BT221T					Biomedical Informatics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand how to acquire, store and use biomedical data.
2. To demonstrate about the general understanding about the computer application in healthcare and biomedicine.
3. Develop knowledge in the ethics that must be followed in Biomedical and health informatics.
4. Describe the functioning of electronic health record systems and management of information in health care organizations.

UNIT I: INTRODUCTION 10 Hrs.

Biomedical Informatics: The Science and the Pragmatics, Biomedical Data: Their Acquisition, Storage, and Use, Biomedical Decision Making: Probabilistic Clinical Reasoning, Cognitive Science and Biomedical Informatics.

UNIT II: COMPUTER ARCHITECTURE AND STANDARDS IN BIOMEDICAL INFORMATICS 10 Hrs.

Computer Architectures for Health Care and Biomedicine, Software Engineering for Health Care and Biomedicine, Standards in Biomedical Informatics, Natural Language Processing in Health Care and Biomedicine.

UNIT III: HEALTH INFORMATION RESOURCES 10 Hrs.

Biomedical Imaging Informatics, Ethics in Biomedical and Health Informatics: Users, Standards, and Outcomes, Biomedical and Health Information Resources, Management of Information in Health Care Organizations, Patient-Centered Care Systems, Public Health Informatics.

UNIT IV: HEALTH CARE MANAGEMENT 10 Hrs.

Telehealth, Patient Monitoring Systems, Imaging Systems in Radiology, Information Retrieval and Digital Libraries, Clinical Decision-Support Systems, Computers in Health Care Education.

40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Classify and Compare the different health records
CO2 : Gain the fundamental knowledge on patient information resources
CO3 : Understand the principles of biomedical informatics
CO4 : Get acquainted with the techniques for the studying and analyzing patient records
CO5 : Focus professionally on ethics in biomedical and health informatics
CO6 : Develop solution of telehealth and Image systems.

TEXT/REFERENCE BOOKS

1. Shortliffe and Cimino, “Biomedical Informatics” 2017, Springer.
2. P. Englebardt, Ramona Nelson, “Health Care Informatics: An Interdisciplinary Approach”, Springer 2021, 2nd edition.